
Counted But Not Centered: Demographic Diversity and Framing in Post-2020 Music Theory Textbooks

Technical Methods Supplement

Full computational methodology for the demographic audit (Part One) and the NLP framing analysis (Part Two)

Purpose of This Document

This supplement documents in full technical detail the methods summarized in the main article, *Counted But Not Centered: Demographic Diversity and Framing in Post-2020 Music Theory Textbooks*. It is intended for readers who wish to evaluate, reproduce, or extend the computational pipeline: the corpus-construction procedure and demographic-coding protocol behind Part One (the demographic audit), the diversity metric applied to each edition, and the three-stage natural language processing (NLP) cascade behind Part Two (the framing analysis), including every model, threshold, and statistical test. The main article presents these methods in lay terms and reports the substantive findings; nothing of analytical substance differs between the two documents.

All databases, extraction scripts, analysis code, and the synthetic corpus needed to reproduce the figures are archived in the project repository. Section numbers and statistical values reported here correspond to those in the main article.

A note on reproducibility and provenance. Every quantity reported in this supplement was computed directly from the study database; no figure is transcribed by hand. The pipeline is deterministic through the first two stages; the third stage invokes a large language model whose outputs, while stable under the fixed decoding settings recorded below, were additionally screened manually (Section 10). Two software dependencies pin specific released versions (`pdfplumber` 0.11.9; the embedding and language models named in Sections 8–9), so that later releases cannot silently alter results.

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Part One: Demographic Audit

Corpus construction, demographic coding, and the distributional diversity metric applied to every edition

1. Corpus Construction

The corpus for the demographic audit comprises ten music theory textbooks, surveyed across twenty editions spanning 2008–2026. Seven are the most widely adopted commercial theory texts in the United States – the same core set examined by Ewell (2019, 2020) and Ewell and Lyons (2024) – and three are open-access texts (Gotham et al. 2023; Hutchinson 2024; Mount 2020) not treated in prior demographic studies. Analyzing multiple editions of the commercial titles, rather than only their most recent printings, is what makes change over time measurable: it isolates what each revision added, removed, or left untouched.

The unit of analysis is the *musical example* – a passage of notated music attributed to a named composer and printed to illustrate a theoretical point. For each edition, the index of musical examples was parsed to enumerate every such example and the composer to whom it is attributed. Where an edition provides no usable index of examples (as with several open-access and born-digital texts), a scraper extracted the same information from the figure captions throughout the body text. Each extracted record pairs a composer identity with the edition and location in which the example appears; repeated appearances of the same composer are retained as distinct tokens, since example *counts*, not merely the roster of names, are the quantity of interest. The resulting database holds 10,387 attributed examples across the twenty editions.

Extraction assistance and verification. A large language model (Claude Sonnet 4.6; Anthropic) assisted with parsing index entries and normalizing the extracted records into tabular form. Every record was subsequently verified and, where necessary, corrected against the source text by the author. The model was used as a transcription aid, not as an arbiter of attribution or demographic category.

2. Demographic Coding Protocol

Each distinct composer was coded on three variables: gender (recorded as M/F), BIPOC status (recorded as Y/N), and birthplace (a place name, subsequently geocoded; see Section 4). Demographic attributes were established by consulting, in fixed priority order, *Oxford Music Online*, the Répertoire International des Sources Musicales (RISM), Wikipedia, and general web search. The priority order is itself a data-quality decision: it privileges scholarly reference works and defers to open sources only when the authoritative ones are silent. Ambiguous or underdetermined cases were coded N/A rather than forced into a category – for instance, Ignatius Sancho, born aboard a transatlantic slave

ship, has no determinate birthplace and is left null on the geographic variable.

2.1. Measurement Caveat

Binary gender and yes/no BIPOC codings treat socially constructed, multidimensional identities as discrete natural kinds. This commits a recognized measurement error: it violates the principle of self-identification and reifies exactly the essentialist distinctions that contemporary social science seeks to interrogate rather than reproduce (Westbrook and Saperstein 2015; Cameron and Stinson 2019; Roth 2016). The justification for retaining these codings here is specific to the object of study. The subjects are, overwhelmingly, historical and long-dead figures, and the records any such audit must rely upon – birth registers, census schedules, administrative rolls – are themselves almost universally binary and observer-assigned. Retrospective self-identification is therefore not a stricter option withheld but a methodological impossibility, and the crude codings function as a documented “least-worst” instrument rather than an endorsed ontology.

3. Distributional Diversity: Pielou Evenness (J')

Counting distinct composers measures *richness* but not *evenness*: an edition can list many composers while concentrating nearly all of its examples on a few. To capture how equitably examples are distributed across the composer pool, this audit adopts Pielou’s evenness index, J' , imported from biodiversity ecology (Pielou 1966). For an edition with S distinct composer identities and relative example shares p_i (the fraction of the edition’s examples attributed to composer i), let Shannon entropy be

$$H' = - \sum_{i=1}^S p_i \ln p_i,$$

and define Pielou evenness as its ratio to the maximum entropy attainable for that richness,

$$J' = \frac{H'}{\ln S}.$$

J' ranges from 0, when a single composer receives every example, to 1, under perfect equipartition across all S composers. Its defining property is *independence from richness*: two editions with very different values of S can share an identical J' if their example shares have the same shape. This is precisely what the audit needs – a statistic that answers whether an edition’s diversity derives from many roughly co-equal voices or from a handful of dominant figures trailed by a long tail of rarely cited names. Reported edition-level J' values, and the pre-/post-2020 aggregate shift from $J' = 0.729$ to $J' = 0.840$, are computed by this definition over each edition’s full example distribution.

Why evenness rather than a concentration index. Diversity indices such as Herfindahl–Hirschman or raw Shannon entropy confound richness with evenness, so a gain can reflect either more composers or a flatter distribution. Normalizing entropy by $\ln S$ separates the two: J' isolates the shape of the distribution, letting the composer count be reported alongside it as an independent quantity. The two together distinguish an edition that adds many peripheral names from one that genuinely redistributes attention.

4. Geographic Coding

Birthplaces were geocoded to latitude/longitude coordinates to support the geographic analyses. Two derived measures follow from the coordinates: the count of distinct countries of origin represented in an edition, and each composer’s

great-circle distance from Vienna, taken as a reference point for the German-speaking common-practice core. Distance-from-Vienna distributions (reported by edition) and the share of composers born beyond 2,000 km from Vienna operationalize geographic breadth without presupposing a center other than the historically dominant one. Composers coded N/A on birthplace are excluded from geographic computations only, and remain in all gender and racial tallies.

Part Two: NLP Framing Analysis

Passage segmentation, the three-stage framing cascade, concept tagging, and the statistical tests behind the reported associations

The framing analysis asks not *who* appears but *how* each composer’s work is positioned within the theoretical exposition. It follows Lucy et al. (2020), who applied NLP to fifteen U.S. history textbooks and showed that representation counts can conceal systematic disparities in how groups are framed. Where that study analyzed language describing historical actors, the present analysis classifies the rhetorical role assigned to each *musical example* – the pedagogical function the composer’s work is made to perform. This part operates on the latest edition of each of the ten textbooks.

5. Text Extraction and Passage Segmentation

Because the ten source texts exist in incompatible formats (print PDFs of varying vintage, born-digital PDFs, HTML, and ePub), no single extraction routine suffices. A bespoke extraction script was written for each source, tuned to its structural peculiarities, but all feed a common downstream representation. For PDF sources, the raw text of each page and the book’s logical structure were reconstructed from layout signals using `pdfplumber` (Singer-Vine 2026, version 0.11.9); for HTML and ePub sources, the corresponding markup tags were used.

The reconstructed text is segmented into *passages*, each a continuous stretch of prose within a single chapter and section. Each passage is assigned a structural *role*:

- **Central** – on-topic body text that carries the section’s analytic argument;
- **Supplementary** – off-topic or structurally non-body material (sidebars, historical asides, front/back matter);
- **Application** – exercises and other material students are asked to work through.

Role assignment is automatic and content-based: it measures how much a passage’s thematic vocabulary overlaps with the concept set of its enclosing chapter, so that a passage on an unrelated topic embedded in a chapter is flagged as supplementary rather than central. For each composer mention, a context window of approximately two sentences on either side is extracted; the framing classifiers operate on this window, concentrating the analysis on the local rhetorical environment of the mention. The framing corpus comprises 6,313 such passages.

6. Framing Categories

Each context window is assigned one of four substantive framing categories, or a residual *neutral* label when no category applies with confidence. The categories are defined by the rhetorical function the surrounding prose assigns to the example, not by any property of the music itself.

Category	Definition and diagnostic sense
Normative	The composer or the work is presented as foundational, standard, or representative of established practice – a natural instance of how a procedure typically works. <i>E.g.</i> , “The following example [by José Maurício Nunes Garcia] shows a pair of dominant seventh chords... resolving in a conventional manner to the tonic” (Mount, §19.4).
Additive	The composer extends or varies a concept already established through other (usually dominant) examples – the “and also...” register, positioned alongside but subordinate to a primary model. <i>E.g.</i> , “This scale [octatonic] was also of interest to Scriabin, Stravinsky, Bartók, Debussy, Messiaen, and countless jazz composers” (Kostka & Almén, p. 470).
Exceptional	The work is flagged as unusual, rare, or markedly different from the surrounding norm; the language marks it as a special case or curiosity. <i>E.g.</i> , “However, Mozart appears to be setting up a brief deception with this unusual cadence” (Kostka & Almén, p. 347).
Corrective	The surrounding language explicitly acknowledges that the example runs contrary to a norm, or names a prior exclusion – an act of recognition. <i>E.g.</i> , “Notice, too, that the melody [by Finney]... is doubled at the octave (a device Schoenberg consciously avoided)” (Kostka & Almén, p. 518).

The corrective category proved extremely sparse in practice (eight instances across the entire corpus) and is therefore dropped from inferential comparisons, which restrict to the three substantive categories that carry adequate support.

7. The Three-Stage Framing Cascade

Framing labels are assigned by a cascade of three classifiers of increasing capability and cost. Each stage is applied only to the windows its predecessors could not confidently label, so that inexpensive, transparent methods resolve the clear cases and expensive computation is reserved for genuinely ambiguous ones. Windows that reach the end of the cascade without a confident assignment retain the neutral label.

7.1. Stage One — Weighted Lexicon plus TF–IDF Similarity

The first stage combines a hand-built lexicon with a distributional measure. For each framing category, a set of weighted indicator terms was curated as regular expressions matching characteristic rhetorical markers, weighted by diagnostic strength – for example, *common practice* and *canonical* weighted heavily toward normative; *exotic* and *unorthodox* toward exceptional; *historically overlooked* and *deserves recognition* toward corrective. These weights encode domain judgment: terms ubiquitous in analytical prose were downweighted or excluded outright (for instance, *different* was omitted as too common to discriminate between categories).

In parallel, each window is scored by term frequency–inverse document frequency (TF–IDF) cosine similarity against five short prototype sentences written to exemplify each category. TF–IDF weights each term by how often it occurs in the window against how broadly it occurs across the corpus, so that distinctive terms count for more than common ones; cosine similarity then measures the angle between the window’s weighted term vector and each prototype vector.

The two signals are combined as

$$\text{score} = \text{lexical score} + \alpha \cdot \text{TF-IDF similarity}, \quad \alpha = 0.30,$$

so that the distributional signal supplements, rather than overrides, the explicit lexical evidence. A window whose strongest combined score falls below a minimal threshold is left neutral and passed to Stage Two.

7.2. Stage Two — Contextual Sentence Embeddings

The second stage replaces surface term matching with a measure of meaning. Each still-neutral window is encoded with a BERT-family sentence-embedding model, BAAI/bge-large-en-v1.5 (Xiao et al. 2023), selected for its strong standing on the Massive Text Embedding Benchmark (Muennighoff et al. 2022). Unlike a bag-of-words representation, such a model encodes each window in the context of its surrounding words – the same contextual-encoding advance that lets a search engine interpret the intent behind a query rather than matching keywords alone. The window embedding is compared by cosine similarity against embedded prototype sentences for each category, and the window is reassigned from neutral to its nearest category only when that similarity meets or exceeds 0.50, a threshold set to admit semantically clear cases while leaving genuinely ambiguous windows for the final stage. As an unsupervised check on label coherence, a k -means clustering of the embeddings is retained and compared against the assigned categories; it is a diagnostic, not a labeling step.

7.3. Stage Three — Zero-Shot Adjudication by a Language Model

The residual windows – those still neutral, or labeled with low embedding confidence – are submitted to an instruction-tuned large language model (Llama-3.1-8B-Instruct; Grattafiori et al. 2024) performing zero-shot classification. The model receives the four category definitions and is asked to assign one, with no corpus-specific training examples supplied. This stage acts as a reasoning-based adjudicator, bringing contextual judgment about what each framing category means in music theory pedagogy that neither the lexical nor the embedding stage can supply. It settled only 1.63% of all framing assignments, and those outputs were screened manually for plausibility.

Pipeline parameters at a glance.

Stage 1 blend weight	$\alpha = 0.30$ (TF-IDF similarity added to lexical score)
Stage 2 embedding model	BAAI/bge-large-en-v1.5 (Xiao et al. 2023)
Stage 2 acceptance threshold	cosine similarity ≥ 0.50 vs. category prototype
Stage 2 diagnostic	k -means clustering of embeddings (coherence check only)
Stage 3 model	Llama-3.1-8B-Instruct (Grattafiori et al. 2024), zero-shot
Stage 3 share of labels	1.63% of all assignments; manually screened
Residual	windows unresolved at all stages $\rightarrow$ <i>neutral</i>

8. Concept Tagging

Independently of framing, each passage is tagged with the music theory concepts it engages. A controlled vocabulary of 248 concepts was compiled from the corpus itself, retaining only concepts attested in ten or more passages so that rare or idiosyncratic terms do not enter the analysis. Passages are matched against this vocabulary using regular expressions with alias folding, which normalizes variant terminology (spelling variants, abbreviations, and synonymous phrasings) to a single canonical concept. The concept tags support the concept-association analysis in Section 11.

9. Model-Generated Labels: Confidence and Screening

Two qualifications attach to every framing figure in the study, and are stated here so that downstream results are read with appropriate caution. First, the labels are *model-generated rather than hand-coded*: they are the output of the cascade above, not of human annotation. Second, classifier confidence varies by category. Mean assignment confidence is highest for the additive label (0.88) and lower for normative (0.70) and exceptional (0.72). Consequently, the additive finding is the best-supported of the three, and exceptional/normative contrasts are treated as weaker evidence throughout. The 1.63% of labels set by the Stage-Three language model were additionally reviewed by hand for plausibility. These confidence figures are reported alongside the substantive results rather than suppressed, in keeping with the study's separation of evidence from interpretation.

10. Statistical Procedures

All inferential claims in the framing analysis rest on the procedures below. Effect sizes are reported alongside p -values throughout, since with a corpus of several thousand passages statistical significance is easy to attain and says little on its own about magnitude.

10.1. Association Between Framing and Group Membership

The primary framing contrast compares dominant and marginalized composers across the three substantive framing categories (corrective dropped for sparsity). A *chi-square test of independence* evaluates whether framing category and group membership are associated beyond chance ($\chi^2 = 23.1$, $df = 2$, $p < .001$). To localize *where* the association lies, *Fisher's exact tests* are run on the relevant two-by-two tables, one framing category against the rest: dominant composers receive normative framing at greater odds than marginalized composers ($OR = 1.34$, $p < .001$), and marginalized composers receive additive framing at greater odds than dominant ones ($OR = 1.47$, $p < .001$). Magnitude is summarized by *Cramér's V*, an effect-size measure for contingency tables scaled from 0 (no association) to 1 (perfect association); here $V = 0.071$, indicating that group membership accounts for only a small share of the variation in framing. The association is thus statistically reliable but modest – a conclusion the effect size makes explicit and the p -value alone would obscure.

10.2. Homogeneity of Framing Across Books

Whether the normative-framing rate for marginalized composers is constant across textbooks is tested by a *chi-square test of homogeneity* over the normative-versus-other split, restricted to the nine books with at least ten marginalized mentions (Aldwell, Schachter, and Cadwallader dropped for insufficient support). The test rejects a common rate ($\chi^2 = 17.9$, $df = 8$, $p = .022$): framing varies with editorial choice book by book rather than being a fixed property of the discipline.

10.3. Structural Placement by Gender and Race

The association between demographic status and structural role (central / supplementary / application) is tested separately on each axis. For gender, $\chi^2 = 57.6$, $df = 2$, $p < .001$, with *Cramér's V* = 0.102; for BIPOC status, $\chi^2 = 46.4$, $df = 2$, $p < .001$, with $V = 0.091$. Both are reliable associations of small-to-moderate magnitude.

10.4. The Integration Index

Structural centrality is summarized per book by an *integration index*: the proportion of passages mentioning at least one marginalized composer that also mention at least one dominant composer. A high value indicates marginalized composers woven into shared analytic company; a low value indicates siloing. The index is computed over marginalized *passages* (not mentions), a distinction that matters when a single passage carries several mentions.

11. Concept Associations with Multiple-Comparison Control

The final analysis asks which music theory concepts disproportionately accompany each group. Passages containing *only* dominant-group composers ($n = 1,168$) are compared with passages containing *only* marginalized composers ($n = 276$); mixed passages are excluded to keep the contrast clean. For each of 185 concepts meeting a minimum support threshold, a *Fisher's exact test* evaluates whether the concept's rate of appearance differs between the two passage sets.

Running 185 simultaneous tests at $\alpha = .05$ would be expected to yield roughly nine false positives by chance alone. To control this, p -values are adjusted with the *Benjamini–Hochberg procedure*, which bounds the *false discovery rate* – the expected proportion of false positives among the results declared significant – rather than the stricter family-wise error rate. This choice suits exploratory screening: it retains genuine effects that a Bonferroni correction would discard while still bounding contamination of the significant set. Twenty-one concepts survive correction, more than double the chance expectation, and it is only these that are interpreted.

Why control the false discovery rate rather than the family-wise error rate. Bonferroni-style family-wise control asks that the probability of *even one* false positive be small, which at 185 tests would suppress most true effects along with the false ones. Benjamini–Hochberg instead limits the *share* of declared findings expected to be false. For a screening analysis whose purpose is to surface candidate concept associations for interpretation – not to certify any single one in isolation – the latter is the appropriate and more powerful standard.

Software and References

Computational environment, dependencies, and works cited

12. Computational Environment and Dependencies

The pipeline is implemented in Python, with figure generation and the diversity and network computations in R. Two dependencies are version-pinned because later releases could alter output: `pdfplumber 0.11.9` (Singer-Vine 2026) for PDF layout reconstruction, and the two named models — the embedding model `BAAI/bge-large-en-v1.5` (Xiao et al. 2023) and the language model `Llama-3.1-8B-Instruct` (Grattafiori et al. 2024). Pielou evenness and the composer co-occurrence network are computed in R. Complete environment specifications, the extraction and analysis scripts, the assembled database, and a synthetic corpus sufficient to exercise the full pipeline are provided in the project repository; the repository is the authoritative record of any implementation detail not restated here.

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